

Amrit Puri (He/Him)

Software Engineer / Educator

Manamaiju, Tarkeshwar-9, Kathmandu 44600, Nepal

+977-9841400296 | amritpuri1@gmail.com

amritpuri.com.np | linkedin.com/in/amwrit | github.com/amwrit

Profile

Senior Software Engineer and Computer Engineer with experience spanning **C++ software development, system-level application engineering, AI integration, automated testing, and programming education**. Currently working on medical ultrasound software, developing and integrating C++ components for image-processing and AI-assisted clinical workflows.

Experienced in translating product and system requirements into maintainable software designs, working across application, algorithm, and hardware/simulation boundaries, and building automated test infrastructure to improve development and validation efficiency.

Alongside software engineering, I have more than a decade of experience in **education, curriculum design, mentoring, and project leadership**, including founding and leading a nonprofit programming education initiative for public-school students. I enjoy working at the intersection of technology, engineering, education, and creative problem-solving.

I value clean architecture, continuous learning, practical experimentation, and building technology that is reliable, understandable, and useful to its users.

Programming Languages & Technologies

- C++, Python, Ruby on Rails, SQL, VB.NET, JavaScript, C#
- Object-Oriented Programming, STL, RAIL, SOLID principles
- Design patterns, modular architecture, debugging, performance optimization
- Image processing, signal processing, classical machine learning
- OpenCV, FFT, DICOM

Software Engineering

- System design and architecture
- Requirements analysis and technical design
- Software Design Specifications (SDS/SRS)
- Test planning and Software Test Procedures (STP)
- Automated testing and test framework development
- Unit, functional, integration, and system-level testing
- Debugging and root-cause analysis
- Legacy application integration and modernization
- Code review and technical mentoring

Testing & Automation

- Python
- PyTest
- CTAF
- Keyword-driven testing

- Test automation
- Test data and golden-data validation
- Simulation and test-environment development

Development Tools & Collaboration

- Git, GitHub, Bitbucket
- JIRA, Confluence, Trello, Notion, Slack
- Scrum, Kanban
- Technical documentation and engineering workflows

Database

- MySQL
- SQLite

Education & Facilitation

- Programming education
- STEAM education
- Curriculum and lesson design
- Project-based learning
- Computational thinking
- Design thinking
- Teacher training and professional development
- Mentoring and learner-centered facilitation

Project & Organizational Management

- Technical/project planning
- Proposal and report writing
- Budget management
- Business Canvas Model
- Business planning
- Donor coordination
- Milestone planning
- Team coordination and mentoring

Creative Skills

- Adobe Photoshop
- Inkscape
- Charcoal and graphite sketching
- Basic photography and videography

Professional Experience

Senior Software Engineer — Leapfrog Technology

Feb 2025 – Present

- Develop and maintain C++ desktop applications for medical ultrasound systems, with emphasis on maintainability, performance, reliability, and clean object-oriented design.
- Apply STL, SOLID, RAII, design patterns, and modular architecture to develop and refactor production software.
- Design and implement software components that integrate AI inference and image/signal-processing algorithms into existing ultrasound application workflows.
- Collaborate with algorithm, application, QA, and other engineering teams to translate algorithm capabilities and product requirements into production-ready software.
- Contribute to feature architecture and technical design, including defining component responsibilities, interfaces, data flow, and integration boundaries.
- Develop and maintain software around B-mode and Doppler ultrasound data, including raw-data loading, frame management, mode-specific processing, and algorithm integration.
- Designed and implemented a modular raw ultrasound data management layer for development and validation workflows, separating clip metadata parsing, data loading, and application-facing interfaces.
- Built application and simulator components to support development and validation of hardware-dependent workflows, reducing reliance on physical ultrasound hardware during software development and testing.
- Contributed to HostDspSimulator and HDS integration workflows, enabling application features to be developed and validated against simulated hardware interfaces.
- Develop automated test infrastructure using Python, PyTest, CTAF, and keyword-driven testing, improving repeatability and reducing manual validation effort.
- Create reusable automated testing templates and test workflows for validating algorithm and application changes against representative ultrasound data.
- Work with golden/reference ultrasound data to validate processing behavior and detect regressions during feature development.
- Investigate complex integration and data-flow issues across application, algorithm, simulator, and hardware abstraction layers using systematic debugging and root-cause analysis.
- Contribute to Software Requirements Specifications, Software Test Procedures, technical design documentation, and other engineering deliverables required for regulated medical-device development.
- Participate in code reviews and provide technical feedback focused on architecture, maintainability, testability, and long-term system impact.
- Support junior engineers through technical discussions, design reviews, debugging, and knowledge sharing.
- Balance feature delivery with architectural quality, testability, and maintainability in a regulated medical-device software environment.

Selected Engineering Contributions

Vascular Flow Assist (VFA) — C++ / Ultrasound / Signal Processing

- Contributed to integration and enhancement of an automated vascular-flow measurement workflow within a legacy ultrasound application.
- Developed application-side components for handling raw B-mode and Doppler ultrasound data, including clip metadata, frame access, and mode-specific information.
- Built a reusable model application from the ground up to load and process raw ultrasound .frm clips for algorithm development and validation.
- Designed a Manager facade that separated application-level data access from underlying clip parsing and loading components.
- Integrated algorithm libraries while keeping client-side application code modular and maintainable.
- Developed automated validation workflows using representative/golden .frm data to identify algorithm and integration issues.
- Supported automated generation and validation of test results through PTAF-based workflows.

Bladder Assist — C++ / AI Integration / Ultrasound

- Contribute to development of an AI-assisted ultrasound workflow for bladder-volume evaluation and post-void residual assessment.

- Work across application and AI-engine boundaries to integrate inferencing functionality into the ultrasound workflow.
- Contribute to intelligent freeze/recall and automated caliper-placement workflows for consistent bladder measurement.
- Develop and integrate components within the existing HostDsp/HDS architecture, including simulator support for hardware-dependent functionality.
- Work with AI engine services and application components while maintaining clear separation between hardware interfaces, service-layer functionality, and UI/application workflows.
- Contribute to technical design, requirements analysis, implementation, debugging, and automated testing throughout the software development lifecycle.

Founder / Project Lead — Code Chautari

Nov 2018 – Dec 2024

- Founded and led **Code Chautari**, a programming education initiative providing free technology and programming education to public-school learners.
 - Secured and managed **USD 15,000+ in annual funding and support** to sustain program operations and educational activities.
 - Led project visioning, strategic planning, budgeting, implementation, reporting, and stakeholder coordination.
 - Designed and delivered programming curricula covering **computer literacy, Google Workspace, computational thinking, Scratch, C, and Python**.
 - Developed hands-on and project-based learning experiences designed for learners with limited prior exposure to programming.
 - Worked directly with public-school learners and teachers to introduce programming and computational thinking in resource-constrained educational environments.
 - Managed donor communication, funding reports, project milestones, and financial records.
 - Coordinated curriculum development and collaborated with educators and mentors to review and improve lesson plans.
 - Established learner progression pathways from foundational computer literacy and Scratch toward text-based programming.
 - Supported multiple cohorts of learners through programming courses and mentorship activities.
 - Provided laptops and learning resources to outstanding learners to support continued independent learning.
 - Built partnerships and secured access to additional learning resources, including extended programming-learning platform access for participating students.
 - Designed and facilitated teacher workshops and programming-learning activities to extend the project's impact beyond individual student cohorts.
 - Managed the project from initial concept through implementation, evaluation, and eventual transition/closure.
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Content Generation Team Lead & STEAM Facilitator — Karkhana

Apr 2017 – Sep 2018

- Led a creative content team responsible for developing **100+ hands-on STEAM learning activities and learning kits**.
- Designed activities around the 4Cs: **Critical Thinking, Creativity, Collaboration, and Communication**.
- Managed content quality, lesson design, activity development, and kit preparation for the BeeCreative program.

- Conducted weekly sprint planning, established priorities, tracked progress against milestones, and coordinated delivery across the team.
 - Mentored team members and supported their professional and personal development.
 - Reviewed and provided feedback on educational content and lesson plans.
 - Collaborated with other teams to coordinate information, resources, and program requirements.
 - Designed and facilitated hands-on STEAM activities for learners.
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Science Teaching Fellow — Teach For Nepal

Feb 2015 – Apr 2017

- Taught secondary-level Science in a public school in Sindhupalchowk, Nepal.
 - Supported **Grade 10 students in achieving a 100% pass rate in SEE Science** during the teaching period.
 - Introduced project-based learning approaches to make Science learning more practical and learner-centered.
 - Organized science exhibitions and student projects to encourage experimentation and applied learning.
 - Managed student learning data and tracked academic progress to identify areas requiring additional support.
 - Initiated community-based educational projects and collaborated with teachers and school leadership.
 - Conducted teacher-learning and professional-development workshops.
 - Developed a **VB.NET student record-management application** to support student data management and academic-result publishing.
 - Participated in school-level project planning and proposal development.
 - Supported discussions around new educational initiatives and contributed to proposal writing.
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Associate Software Engineer — CloudFactory

May 2013 – Jan 2015

Earlier positions: Programming Intern / Junior Software Engineer, 2013–2014

- Developed new features, fixed bugs, and optimized existing functionality in **Ruby on Rails web applications**.
 - Worked with designers to translate UI wireframes into functional web applications.
 - Developed and maintained unit, functional, acceptance, and performance tests.
 - Investigated and resolved software defects through structured debugging and testing.
 - Worked with Ruby on Rails, JavaScript, CSS, SQL, and related web technologies.
 - Tested, debugged, and deployed applications into production environments.
 - Participated in code reviews and collaborative development workflows.
 - Mentored trainees and junior software engineers, supporting their technical development and confidence.
 - Continued independent learning in software development, web technologies, testing, and application architecture.
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Other Professional Experiences

Educational Consultant — DP World

Nov 2024 – Jan 2025

- Contributed educational and technology expertise to project development and learning-related initiatives.

C Programming Module Facilitator — King's College

Sep 2024 – Feb 2025

- Designed and facilitated C programming learning sessions for students.
 - Supported learners in developing foundational programming concepts and problem-solving skills.
 - Used practical exercises and hands-on programming activities to reinforce concepts.
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Major Projects

VFA – C++ / Ultrasound Signal Processing – *LF Technology* | 2024

- Contributed in enhancing the VFA module for a desktop application for ultrasound signal processing.
- Designed modular classes for raw data reading, frame management, and algorithm integration, supporting B-mode and Doppler modes.
- Implemented automated testing templates to validate algorithm updates and detect potential issues, maintaining high QA standards.
- Streamlined interaction with algorithm libraries while keeping the client-side code minimal and maintainable.

Audio Signal Analyzer – C++ / FFT / Win32 – *Independent Project* | 2024

- Developed a desktop application for processing uncompressed 8-bit WAV audio.
- Implemented Fast Fourier Transform (FFT) processing to convert audio from time-domain to frequency-domain representations.
- Designed modular object-oriented components for audio loading, FFT processing, and instrument-specific analysis.
- Implemented experimental analysis for instruments including guitar and bass.
- Focused on clean architecture, modularity, maintainability, and straightforward library setup.

Body Part Recognition – C++ / OpenCV / SVM / KNN | *Independent Project* | 2023

- Built a computer-vision classification system for identifying human body parts from DICOM images.
- Implemented preprocessing and feature-extraction workflows.
- Applied classical machine-learning techniques including Support Vector Machines (SVM) and K-Nearest Neighbors (KNN).
- Experimented with feature selection and classification approaches to improve recognition accuracy.

L&F – Lost and Found – Python / Flask / SQLite – *Harvard CS50x Final Project* | 2020

- Developed a web application for registering, searching, and claiming lost items.
- Implemented user authentication, database integration, and responsive web interfaces.
- Applied full-stack development practices with an emphasis on usability and reliability.

Ed-Dy Run – Unity / C# – *Harvard CS50G Project* | 2020

- Developed a 2D educational game combining entertainment with problem-solving challenges.
- Implemented gameplay logic and physics-based mechanics using Unity and C#.
- Applied interactive-learning and game-design principles to create an engaging educational experience.

ASMAS – Academic Student Management and Analysis System – *Independent / Educational Project* | 2016–2017

- Built a VB.net desktop application for managing student databases and publishing academic results for public schools.
- Integrated user-friendly GUI, automated report generation, and secure data handling.
- Deployed and actively used in classroom settings for efficient student management.

Other Programming Projects

- [CS50 AI Projects \(Python\)](#) — Implementations and assignments from Harvard's CS50 Artificial Intelligence course.
 - [CS50G Projects](#) (Lua and Unity) — Game-development projects completed as part of Harvard's CS50 Game Development course.
 - **yourGUIDE** — Navigation website project.
 - **BillsYS** — Desktop billing system application.
 - **infoKU** — Android application developed for university students.
 - **Morphological Analyzer:** University IT project involving language/text processing.
 - Portfolio: github.com/amwrit
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Certifications

HarvardX (CS50 Series)

- Introduction to Computer Science
- Game Development
- Artificial Intelligence
- Programming with Scratch
- Programming with Python
- Databases with SQL
- CS50 for Educators Conference

Teach For All — Global Girls' Education Course

Kathmandu University — Academic Writing for Graduate Students

Events & Workshops

- **Jan 2025** — Speaker & Workshop Facilitator, *NDC London*, United Kingdom
- **Oct 2023** — Speaker & Workshop Facilitator, *NDC Oslo*, Norway
- **Feb 2023** — *Scratch Learning Philosophy Workshop*, India
- **Oct 2019** — Speaker & Facilitator, *NDC Sydney*, Australia
- **Sep 2019** — Facilitator, *Scratch It: Girls Learn to Code*, Nepal
- **Aug 2019** — Participant, *Teach For All Asia-Pacific Alumni Gathering*, Kuala Lumpur
- **Sep 2018** — Participant, *Asia-Pacific Educators Gathering*, Bangkok
- **Jul 2018** — Facilitator, *Summer Camp*, Lahore, Pakistan
- **Feb 2018** — Participant, *BioCamp Tokyo: Gardens as Biotechnik*, Japan Foundation Asia Center

- **Dec 2017** — Presenter, *Teachers Together*, Nepal
 - **Sep 2017** — Selected, *MIT Media Lab BioSummit 2018*, USA
 - **Feb 2015** — Participant, *Teaching as Leadership Training*, Teach for Nepal
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Awards & Honors

- **Academic Scholarships** — Received scholarships at multiple stages from Grade 6 through Master's level, including Motherland Higher Secondary School, Himalayan White House International College, Kathmandu University, University Grants Commission, and Open Institute for Social Science.
 - **2013** — *First Place*, Threadpaints “Butta” T-Shirt Design Contest
 - **IT Meet Awards**, Kathmandu University — Projects *yourGUIDE & Morphological Analyzer*
 - Multiple **art and photography** competition awards, Kathmandu University
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Education

Master's in Technical and Vocational Education and Training

Kathmandu University (2021–2023) — CGPA: 3.89/4

Higher Post Graduate Diploma in Writing and Research

The Open Institute for Social Science (2021–2022)

Bachelor's in Computer Engineering

Kathmandu University (2009–2013) — CGPA: 3.52/4

HSEB +2 in Science

Himalayan White House International College — 74.1%

S.L.C

Motherland Higher Secondary School — 87.5%

Art & Design

- Portfolio: behance.net/amwritpuri
- Art & Design: fb.com/ArtainCreations
- Charcoal and graphite sketching
- Graphic design
- Photography
- Basic videography